The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Zero-Hidden Spectral Generator Law
Necessity / Uniqueness from Zero-Hidden Stationary Hessian Law

Abstract

The current benchmark-facing theorem on the active primitive-reservoir observer-reduction line already sharpened the burden once more: the zero-hidden selector generator law is no longer merely available, because it is derived canonically from a zero-hidden spectral generator law and is necessary there up to the obvious zero/hidden spectral identifications. The remaining honest task is therefore no longer to justify the selector-law layer. It is to justify why the spectral law itself should hold.

The present note takes the smallest honest step now available below that spectral layer without claiming final microphysical closure. Rather than postulate a positive self-adjoint spectral generator directly, it collapses the spectral package into a smaller zero-hidden stationary Hessian law. For each sector this deeper package consists of a compact convex stationary state body, affine observer-data and shell-response readouts, a nonnegative stationary potential with an exact zero-hidden stationary locus, exact body splitting over zero-hidden and hidden directions, a common zero-locus Hessian that vanishes on zero directions and is coercively positive on hidden directions, symmetry and time-reversal actions preserving the full stationary law, an exact zero-response shell realization on the zero locus, fixed-data shell solvability on that same locus, and a stationary-Hessian coercivity inequality on data-invisible variations modulo the inert zero-hidden kernel.

From that stationary Hessian law one derives canonically every constituent of the spectral package that now requires justification: the compact convex spectral state body, the affine observer-data and shell-response readouts, the positive self-adjoint spectral generator, the exact zero spectral band and positively gapped hidden spectral band, the induced zero/hidden splitting and hidden body, the symmetry and time-reversal structure, the exact zero-response shell realization on the zero band, fixed-data shell solvability on that locus, the inert zero-hidden kernel and projector, and the spectral coercivity inequality. Within this stationary-Hessian class those spectral constituents are not merely available. They are necessary and unique up to the obvious orthogonal identification preserving the zero/hidden splitting and the recorded data and shell coordinates. The benchmark package remains fixed throughout: the one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has again narrowed what counts as an honest next step. The primitive-observer-reduction datum theorem, the defect-fiber factorization theorem, the one-metric output theorem, the chain-forcing theorem, the selector

theorem, the stationary-construction theorem, the explicit-package theorem, the primitive-normal-form theorem, the ambient-shell-law theorem, the combined-response theorem, the graph-collapse theorem, the completion-core theorem, the ambient-dynamics theorem, and the later selector-law theorem each discharged what was honestly open at its own layer. The latest result matters because it removes another tempting stopping point. Once the selector-law package is itself a canonical reduction of a deeper spectral law, it is no longer enough to revisit the selector layer under a different presentation.

The remaining honest burden is therefore one layer deeper again. The task is now to explain why the zero-hidden spectral generator law should exist, rather than letting the spectral state body, the data and shell readouts, the positive self-adjoint generator, the zero/hidden splitting, the hidden gap, the shell realization, the solvability statement, and the spectral coercivity inequality stand as the new deepest disciplined assumptions. That is the burden addressed here.

The present note takes the smallest honest collapse presently available on the active line. It does not claim a final microscopic substrate law. Its gain is narrower and more upstream. It replaces the primitive spectral generator by a smaller stationary law: one nonnegative stationary potential whose common zero-locus Hessian canonically generates the spectral operator together with the hidden-band coercivity. In that precise sense, the spectral layer itself ceases to be an independent stopping point.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not claim that final substrate microphysics has already been found. Its narrower theorem is only that, on the active primitive-observer-reduction line, the zero-hidden spectral generator law is itself forced by a smaller zero-hidden stationary Hessian law, and is unique there up to the obvious orthogonal identification preserving the zero/hidden splitting.

2. Fixed Primitive Continuation Data

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}},$$

a visible reduction map

$$\mathcal{R}_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \to X_\sigma,$$

and shell-level current and equilibrium carriers

$$\mathfrak{C}_\sigma^{\mathrm{cur}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{cur}}, \qquad \mathfrak{C}_\sigma^{\mathrm{eq}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{eq}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *Fixed shell extractors*

$$\mathscr{E}_\sigma^{\mathrm{cur}} : Z_\sigma^{\mathrm{cur}} \to Y_\sigma^{\mathrm{cur}}, \qquad \mathscr{E}_\sigma^{\mathrm{eq}} : Z_\sigma^{\mathrm{eq}} \to Y_\sigma^{\mathrm{eq}},$$

together with the recorded shell composites

$$\mathcal{J}_\sigma^{\mathrm{prim}} = \mathscr{E}_\sigma^{\mathrm{cur}} \circ \mathfrak{C}_\sigma^{\mathrm{cur}}, \qquad \Theta_\sigma^{\mathrm{prim}} = \mathscr{E}_\sigma^{\mathrm{eq}} \circ \mathfrak{C}_\sigma^{\mathrm{eq}}. \tag{1}$$

4. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{2}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$.

Remark 1. *Definition 1 is not reopened here. The primitive shell data and the benchmark one-metric / ordinary-matter outputs remain fixed continuation data. The work begins one layer deeper again: why the spectral law that now carries the active line should itself be forced.*

3. Target Spectral Generator Law

Definition 2 (Zero-hidden spectral generator law)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* zero-hidden spectral generator law *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean spectral state space*

$$\mathcal{U}_\sigma^{\mathrm{sp}},$$

the observer-data space

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \equiv \widehat{X}_\sigma \times Z_\sigma^{\mathrm{cur}} \times Z_\sigma^{\mathrm{eq}},$$

an ambient shell-response space

$$E_\sigma^{\mathrm{amb}},$$

and a compact convex spectral state body

$$\mathcal{B}_\sigma^{\mathrm{sp}} \subset \mathcal{U}_\sigma^{\mathrm{sp}}$$

with nonempty interior.

2. *An affine observer-data readout*

$$\Lambda_\sigma^{\mathrm{sp,dat}} : \mathcal{U}_\sigma^{\mathrm{sp}} \to \mathcal{Y}_\sigma^{\mathrm{obs}}$$

and an affine shell-response readout

$$\Lambda_\sigma^{\mathrm{sp,sh}} : \mathcal{U}_\sigma^{\mathrm{sp}} \to E_\sigma^{\mathrm{amb}}.$$

3. *A positive self-adjoint spectral generator*

$$\mathcal{L}_\sigma^{\mathrm{sp}} : \mathcal{U}_\sigma^{\mathrm{sp}} \to \mathcal{U}_\sigma^{\mathrm{sp}}, \qquad \mathcal{L}_\sigma^{\mathrm{sp}*} = \mathcal{L}_\sigma^{\mathrm{sp}}, \qquad \langle u, \mathcal{L}_\sigma^{\mathrm{sp}} u\rangle \geq 0$$

for all $u \in \mathcal{U}_\sigma^{\mathrm{sp}}$*. Define the zero-hidden spectral space*

$$V_\sigma^{\mathrm{sp},0} \equiv \ker \mathcal{L}_\sigma^{\mathrm{sp}},$$

the hidden spectral space

$$H_\sigma^{\mathrm{sp}} \equiv (V_\sigma^{\mathrm{sp},0})^\perp,$$

and the orthogonal hidden spectral projector

$$\Pi_\sigma^{\mathrm{sp,hid}} : \mathcal{U}_\sigma^{\mathrm{sp}} \to H_\sigma^{\mathrm{sp}}.$$

4. *A compact convex hidden spectral body*

$$D_\sigma^{\mathrm{sp}} \subset H_\sigma^{\mathrm{sp}}$$

with

$$0 \in \operatorname{int} D_\sigma^{\mathrm{sp}},$$

such that the spectral state body splits exactly as

$$\mathcal{B}_\sigma^{\mathrm{sp}} = \left\{ z + \eta : z \in \mathcal{B}_\sigma^{\mathrm{sp}} \cap V_\sigma^{\mathrm{sp},0}, \quad \eta \in D_\sigma^{\mathrm{sp}} \right\}. \tag{3}$$

5. *An exact hidden-band gap: there exists* $\lambda_\sigma^{\mathrm{sp}} > 0$ *such that*

$$\langle \eta, \mathcal{L}_\sigma^{\mathrm{sp}} \eta\rangle \geq \lambda_\sigma^{\mathrm{sp}} \|\eta\|^2 \qquad \textit{for all } \eta \in H_\sigma^{\mathrm{sp}}. \tag{4}$$

6. *A compact symmetry group*

$$\mathbb{U}_\sigma \subset O(\mathcal{U}_\sigma^{\mathrm{sp}}) \times O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}})$$

and an orthogonal involution

$$\vartheta_\sigma^{\mathrm{sp}} \in O(\mathcal{U}_\sigma^{\mathrm{sp}}) \times O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}})$$

preserving $\mathcal{B}_\sigma^{\mathrm{sp}}$*,* $\Lambda_\sigma^{\mathrm{sp,dat}}$*,* $\Lambda_\sigma^{\mathrm{sp,sh}}$*, and* $\mathcal{L}_\sigma^{\mathrm{sp}}$*.*

7. *A smooth embedding*

$$\jmath_\sigma : X_\sigma \hookrightarrow \widehat{X}_\sigma$$

and a smooth zero-response shell realization

$$\zeta_\sigma^{\mathrm{sp}} : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{U}_\sigma^{\mathrm{sp}}$$

whose image is exactly

$$\mathcal{B}_\sigma^{\mathrm{sp}} \cap V_\sigma^{\mathrm{sp},0} \cap \big(\Lambda_\sigma^{\mathrm{sp,sh}}\big)^{-1}(0),$$

and such that

$$\Lambda_\sigma^{\mathrm{sp,dat}}(\zeta_\sigma^{\mathrm{sp}}(w)) = \big(\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)\big) \tag{5}$$

for every $w \in \mathcal{I}_\sigma^{\mathrm{for}}$.

8. *Fixed-data shell solvability on the zero-hidden spectral space:*

$$\Lambda_\sigma^{\mathrm{sp,dat}}(\mathcal{B}_\sigma^{\mathrm{sp}} \cap V_\sigma^{\mathrm{sp},0}) = \Lambda_\sigma^{\mathrm{sp,dat}}\Big(\mathcal{B}_\sigma^{\mathrm{sp}} \cap V_\sigma^{\mathrm{sp},0} \cap \big(\Lambda_\sigma^{\mathrm{sp,sh}}\big)^{-1}(0)\Big). \tag{6}$$

9. *The inert zero-hidden kernel*

$$K_\sigma^0 \equiv \ker\big(D\Lambda_\sigma^{\mathrm{sp,dat}}|_{V_\sigma^{\mathrm{sp},0}}\big) \cap \ker\big(D\Lambda_\sigma^{\mathrm{sp,sh}}|_{V_\sigma^{\mathrm{sp},0}}\big)$$

and the orthogonal projector

$$P_\sigma^{\mathrm{co}} : V_\sigma^{\mathrm{sp},0} \to K_\sigma^{0\perp} \cap V_\sigma^{\mathrm{sp},0}.$$

10. *Spectral coercivity on data-invisible directions: there exists* $\kappa_\sigma^{\mathrm{sp}} > 0$ *such that for every*

$$w \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad \delta u \in T_{\zeta_\sigma^{\mathrm{sp}}(w)}\mathcal{U}_\sigma^{\mathrm{sp}} = \mathcal{U}_\sigma^{\mathrm{sp}}$$

satisfying

$$D\Lambda_\sigma^{\mathrm{sp,dat}}\,\delta u = 0,$$

one has

$$\|D\Lambda_\sigma^{\mathrm{sp,sh}}\,\delta u\|^2 + \langle \delta u, \mathcal{L}_\sigma^{\mathrm{sp}}\delta u\rangle \geq \kappa_\sigma^{\mathrm{sp}}\left(\|P_\sigma^{\mathrm{co}}((I - \Pi_\sigma^{\mathrm{sp,hid}})\delta u)\|^2 + \|\Pi_\sigma^{\mathrm{sp,hid}}\delta u\|^2\right). \tag{7}$$

Remark 2. *Definition 2 records exactly the spectral-layer objects that now require deeper justification. The present note does not enlarge that package. It asks whether those same objects can be forced by a smaller stationary law beneath them.*

4. Zero-Hidden Stationary Hessian Law

Definition 3 (Zero-hidden stationary Hessian law)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* zero-hidden stationary Hessian law *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean stationary state space*

$$\mathcal{U}_\sigma^{\mathrm{st}},$$

the observer-data space

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \equiv \widehat{X}_\sigma \times Z_\sigma^{\mathrm{cur}} \times Z_\sigma^{\mathrm{eq}},$$

an ambient shell-response space

$$E_\sigma^{\mathrm{amb}},$$

and a compact convex stationary state body

$$\mathcal{B}_\sigma^{\mathrm{st}} \subset \mathcal{U}_\sigma^{\mathrm{st}}$$

with nonempty interior.

2. *An affine observer-data readout*

$$\Lambda_\sigma^{\mathrm{st,dat}} : \mathcal{U}_\sigma^{\mathrm{st}} \to \mathcal{Y}_\sigma^{\mathrm{obs}}$$

and an affine shell-response readout

$$\Lambda_\sigma^{\mathrm{st,sh}} : \mathcal{U}_\sigma^{\mathrm{st}} \to E_\sigma^{\mathrm{amb}}.$$

3. *A smooth nonnegative stationary potential*

$$\Phi_\sigma^{\mathrm{st}} : \mathcal{U}_\sigma^{\mathrm{st}} \to [0, \infty)$$

whose exact zero-hidden stationary locus

$$V_\sigma^{\mathrm{st},0} \equiv \left\{ u \in \mathcal{U}_\sigma^{\mathrm{st}} : \Phi_\sigma^{\mathrm{st}}(u) = 0 \right\}$$

is a linear subspace. Define the hidden stationary space

$$H_\sigma^{\mathrm{st}} \equiv (V_\sigma^{\mathrm{st},0})^\perp,$$

and let

$$\Pi_\sigma^{\mathrm{st,hid}} : \mathcal{U}_\sigma^{\mathrm{st}} \to H_\sigma^{\mathrm{st}}$$

denote the orthogonal projector.

4. *A compact convex hidden stationary body*

$$D_\sigma^{\mathrm{st}} \subset H_\sigma^{\mathrm{st}}$$

with

$$0 \in \operatorname{int} D_\sigma^{\mathrm{st}},$$

such that the stationary state body splits exactly as

$$\mathcal{B}_\sigma^{\mathrm{st}} = \left\{ z + \eta : z \in \mathcal{B}_\sigma^{\mathrm{st}} \cap V_\sigma^{\mathrm{st},0}, \quad \eta \in D_\sigma^{\mathrm{st}} \right\}. \tag{8}$$

5. *A symmetric bilinear form*

$$\mathfrak{H}_\sigma^{\mathrm{st}} : \mathcal{U}_\sigma^{\mathrm{st}} \times \mathcal{U}_\sigma^{\mathrm{st}} \to \mathbb{R}$$

such that for every $z \in V_\sigma^{\mathrm{st},0}$,

$$D\Phi_\sigma^{\mathrm{st}}(z) = 0, \qquad D^2\Phi_\sigma^{\mathrm{st}}(z)[u, v] = \mathfrak{H}_\sigma^{\mathrm{st}}(u, v) \tag{9}$$

for all $u, v \in \mathcal{U}_\sigma^{\mathrm{st}}$. Assume moreover that

$$\mathfrak{H}_\sigma^{\mathrm{st}}(\delta z, u) = 0 \qquad \textit{for all } \delta z \in V_\sigma^{\mathrm{st},0},\ u \in \mathcal{U}_\sigma^{\mathrm{st}}, \tag{10}$$

and that there exists $\lambda_\sigma^{\mathrm{st}} > 0$ *with*

$$\mathfrak{H}_\sigma^{\mathrm{st}}(\eta,\eta) \geq \lambda_\sigma^{\mathrm{st}} \|\eta\|^2 \qquad \text{for all } \eta \in H_\sigma^{\mathrm{st}}. \tag{11}$$

6. *A compact symmetry group*

$$\mathbb{T}_\sigma \subset O(\mathcal{U}_\sigma^{\mathrm{st}}) \times O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}})$$

and an orthogonal involution

$$\vartheta_\sigma^{\mathrm{st}} \in O(\mathcal{U}_\sigma^{\mathrm{st}}) \times O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}})$$

preserving $\mathcal{B}_\sigma^{\mathrm{st}}$, $\Lambda_\sigma^{\mathrm{st,dat}}$, $\Lambda_\sigma^{\mathrm{st,sh}}$, *and* $\Phi_\sigma^{\mathrm{st}}$.

7. *A smooth embedding*

$$\jmath_\sigma : X_\sigma \hookrightarrow \widehat{X}_\sigma$$

and a smooth zero-response shell realization

$$\zeta_\sigma^{\mathrm{st}} : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{U}_\sigma^{\mathrm{st}}$$

whose image is exactly

$$\mathcal{B}_\sigma^{\mathrm{st}} \cap V_\sigma^{\mathrm{st},0} \cap (\Lambda_\sigma^{\mathrm{st,sh}})^{-1}(0),$$

and such that

$$\Lambda_\sigma^{\mathrm{st,dat}}(\zeta_\sigma^{\mathrm{st}}(w)) = (\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)) \tag{12}$$

for every $w \in \mathcal{I}_\sigma^{\mathrm{for}}$.

8. *Fixed-data shell solvability on the zero-hidden stationary locus:*

$$\Lambda_\sigma^{\mathrm{st,dat}}(\mathcal{B}_\sigma^{\mathrm{st}} \cap V_\sigma^{\mathrm{st},0}) = \Lambda_\sigma^{\mathrm{st,dat}}\Big(\mathcal{B}_\sigma^{\mathrm{st}} \cap V_\sigma^{\mathrm{st},0} \cap (\Lambda_\sigma^{\mathrm{st,sh}})^{-1}(0)\Big). \tag{13}$$

9. *The inert zero-hidden kernel*

$$K_\sigma^0 \equiv \ker\big(D\Lambda_\sigma^{\mathrm{st,dat}}|_{V_\sigma^{\mathrm{st},0}}\big) \cap \ker\big(D\Lambda_\sigma^{\mathrm{st,sh}}|_{V_\sigma^{\mathrm{st},0}}\big)$$

and the orthogonal projector

$$P_\sigma^{\mathrm{co}} : V_\sigma^{\mathrm{st},0} \to K_\sigma^{0\perp} \cap V_\sigma^{\mathrm{st},0}.$$

10. *Stationary-Hessian coercivity on data-invisible directions: there exists* $\kappa_\sigma^{\mathrm{st}} > 0$ *such that for every*

$$w \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad \delta u \in T_{\zeta_\sigma^{\mathrm{st}}(w)}\mathcal{U}_\sigma^{\mathrm{st}} = \mathcal{U}_\sigma^{\mathrm{st}}$$

satisfying

$$D\Lambda_\sigma^{\mathrm{st,dat}}\,\delta u = 0,$$

one has

$$\|D\Lambda_\sigma^{\mathrm{st,sh}}\,\delta u\|^2 + \mathfrak{H}_\sigma^{\mathrm{st}}(\delta u,\delta u) \geq \kappa_\sigma^{\mathrm{st}}\left(\|P_\sigma^{\mathrm{co}}((I-\Pi_\sigma^{\mathrm{st,hid}})\delta u)\|^2 + \|\Pi_\sigma^{\mathrm{st,hid}}\delta u\|^2\right). \tag{14}$$

Remark 3. *Definition 3 is smaller than Definition 2 in exactly the way now needed. The positive self-adjoint spectral generator is not stipulated independently. It is extracted from the common zero-locus Hessian of one nonnegative stationary potential, while the hidden gap is the coercive hidden-sector part of that same Hessian.*

5. Canonical Spectral Law Induced by the Stationary Hessian

Definition 4 (Canonical spectral law induced by the stationary Hessian)**.** *Assume Definition 3. Define the canonical zero-hidden spectral generator law by:*

1. $$\mathcal{U}_\sigma^{\mathrm{sp}} \equiv \mathcal{U}_\sigma^{\mathrm{st}}, \qquad \mathcal{B}_\sigma^{\mathrm{sp}} \equiv \mathcal{B}_\sigma^{\mathrm{st}}, \qquad \Lambda_\sigma^{\mathrm{sp,dat}} \equiv \Lambda_\sigma^{\mathrm{st,dat}}, \qquad \Lambda_\sigma^{\mathrm{sp,sh}} \equiv \Lambda_\sigma^{\mathrm{st,sh}}.$$

2. *The spectral generator $\mathcal{L}_\sigma^{\mathrm{sp}}$ is the unique linear map satisfying*
$$\langle u, \mathcal{L}_\sigma^{\mathrm{sp}} v\rangle = \mathfrak{H}_\sigma^{\mathrm{st}}(u,v) \qquad \textit{for all } u,v \in \mathcal{U}_\sigma^{\mathrm{sp}}. \tag{15}$$

3. $$V_\sigma^{\mathrm{sp},0} \equiv V_\sigma^{\mathrm{st},0}, \qquad H_\sigma^{\mathrm{sp}} \equiv H_\sigma^{\mathrm{st}}, \qquad \Pi_\sigma^{\mathrm{sp,hid}} \equiv \Pi_\sigma^{\mathrm{st,hid}}, \qquad D_\sigma^{\mathrm{sp}} \equiv D_\sigma^{\mathrm{st}}.$$

4. $$\mathbb{U}_\sigma \equiv \mathbb{T}_\sigma, \qquad \vartheta_\sigma^{\mathrm{sp}} \equiv \vartheta_\sigma^{\mathrm{st}}, \qquad \zeta_\sigma^{\mathrm{sp}} \equiv \zeta_\sigma^{\mathrm{st}}.$$

Lemma 1 (The stationary Hessian determines a unique spectral generator)**.** *Assume Definition 3. Then there exists a unique self-adjoint positive semidefinite operator*

$$\mathcal{L}_\sigma^{\mathrm{sp}} : \mathcal{U}_\sigma^{\mathrm{sp}} \to \mathcal{U}_\sigma^{\mathrm{sp}}$$

satisfying (15)*. Moreover,*

$$\ker \mathcal{L}_\sigma^{\mathrm{sp}} = V_\sigma^{\mathrm{st},0}, \qquad \operatorname{im} \mathcal{L}_\sigma^{\mathrm{sp}} = H_\sigma^{\mathrm{st}},$$

and

$$\langle \eta, \mathcal{L}_\sigma^{\mathrm{sp}} \eta\rangle = \mathfrak{H}_\sigma^{\mathrm{st}}(\eta,\eta) \geq \lambda_\sigma^{\mathrm{st}} \|\eta\|^2 \qquad \textit{for all } \eta \in H_\sigma^{\mathrm{st}}.$$

Proof. Because $\mathfrak{H}_\sigma^{\mathrm{st}}$ is a symmetric bilinear form on the finite-dimensional Euclidean space $\mathcal{U}_\sigma^{\mathrm{sp}} = \mathcal{U}_\sigma^{\mathrm{st}}$, the Riesz representation theorem yields a unique self-adjoint operator $\mathcal{L}_\sigma^{\mathrm{sp}}$ satisfying (15). Positivity follows from (10) and (11): if $u = z + \eta$ with $z \in V_\sigma^{\mathrm{st},0}$ and $\eta \in H_\sigma^{\mathrm{st}}$, then

$$\langle u, \mathcal{L}_\sigma^{\mathrm{sp}} u\rangle = \mathfrak{H}_\sigma^{\mathrm{st}}(u,u) = \mathfrak{H}_\sigma^{\mathrm{st}}(\eta,\eta) \geq 0.$$

If $z \in V_\sigma^{\mathrm{st},0}$, then (10) gives

$$\langle u, \mathcal{L}_\sigma^{\mathrm{sp}} z\rangle = \mathfrak{H}_\sigma^{\mathrm{st}}(u,z) = 0 \qquad \text{for all } u,$$

so $z \in \ker \mathcal{L}_\sigma^{\mathrm{sp}}$. Conversely, if $u = z + \eta \in \ker \mathcal{L}_\sigma^{\mathrm{sp}}$ with $z \in V_\sigma^{\mathrm{st},0}$ and $\eta \in H_\sigma^{\mathrm{st}}$, then

$$0 = \langle u, \mathcal{L}_\sigma^{\mathrm{sp}} u\rangle = \mathfrak{H}_\sigma^{\mathrm{st}}(\eta,\eta) \geq \lambda_\sigma^{\mathrm{st}} \|\eta\|^2,$$

hence $\eta = 0$ and $u = z \in V_\sigma^{\mathrm{st},0}$. Thus $\ker \mathcal{L}_\sigma^{\mathrm{sp}} = V_\sigma^{\mathrm{st},0}$. Since $\mathcal{L}_\sigma^{\mathrm{sp}}$ is self-adjoint,

$$\operatorname{im} \mathcal{L}_\sigma^{\mathrm{sp}} = \left(\ker \mathcal{L}_\sigma^{\mathrm{sp}}\right)^\perp = H_\sigma^{\mathrm{st}}.$$

The hidden-sector lower bound is exactly (11). □

Proposition 1 (The stationary Hessian law induces a zero-hidden spectral generator law)**.** *Assume Definitions 3 and 4. Then the canonical data of Definition 4 satisfy every requirement of Definition 2.*

Proof. Items (1) and (2) of Definition 2 are immediate from Definition 4. By Lemma 1, $\mathcal{L}_\sigma^{\mathrm{sp}}$ is positive and self-adjoint with

$$V_\sigma^{\mathrm{sp},0} = \ker \mathcal{L}_\sigma^{\mathrm{sp}} = V_\sigma^{\mathrm{st},0}, \qquad H_\sigma^{\mathrm{sp}} = \left(V_\sigma^{\mathrm{sp},0}\right)^\perp = H_\sigma^{\mathrm{st}},$$

and the orthogonal projector onto H_σ^{sp} is exactly $\Pi_\sigma^{\mathrm{sp,hid}} = \Pi_\sigma^{\mathrm{st,hid}}$. This gives item (3).

The compact convex hidden spectral body is $D_\sigma^{\mathrm{sp}} = D_\sigma^{\mathrm{st}}$, and the spectral body splitting (3) is exactly the stationary body splitting (8). This gives item (4). The exact hidden-band gap (4) follows from Lemma 1 with $\lambda_\sigma^{\mathrm{sp}} = \lambda_\sigma^{\mathrm{st}}$, giving item (5).

The symmetry and time-reversal requirements follow because every element of $\mathbb{T}_\sigma$ and $\vartheta_\sigma^{\mathrm{st}}$ preserves $\Phi_\sigma^{\mathrm{st}}$, hence preserves the zero locus $V_\sigma^{\mathrm{st},0}$ and its common Hessian $\mathfrak{H}_\sigma^{\mathrm{st}}$. By the Riesz characterization (15), this also preserves $\mathcal{L}_\sigma^{\mathrm{sp}}$. Thus item (6) holds.

The shell realization and data compatibility in item (7) are exactly (12), because

$$\mathcal{B}_\sigma^{\mathrm{sp}} \cap V_\sigma^{\mathrm{sp},0} \cap \left(\Lambda_\sigma^{\mathrm{sp,sh}}\right)^{-1}(0) = \mathcal{B}_\sigma^{\mathrm{st}} \cap V_\sigma^{\mathrm{st},0} \cap \left(\Lambda_\sigma^{\mathrm{st,sh}}\right)^{-1}(0).$$

Fixed-data shell solvability (6) is exactly (13). The inert zero-hidden kernel and orthogonal projector in item (9) are defined by the same formulas, since

$$\Lambda_\sigma^{\mathrm{sp,dat}} = \Lambda_\sigma^{\mathrm{st,dat}}, \qquad \Lambda_\sigma^{\mathrm{sp,sh}} = \Lambda_\sigma^{\mathrm{st,sh}}, \qquad V_\sigma^{\mathrm{sp},0} = V_\sigma^{\mathrm{st},0}.$$

For item (10), let

$$w \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad \delta u \in T_{\zeta_\sigma^{\mathrm{sp}}(w)}\mathcal{U}_\sigma^{\mathrm{sp}} = \mathcal{U}_\sigma^{\mathrm{st}}$$

with $D\Lambda_\sigma^{\mathrm{sp,dat}}\,\delta u = 0$. Then (15) gives

$$\langle \delta u, \mathcal{L}_\sigma^{\mathrm{sp}} \delta u \rangle = \mathfrak{H}_\sigma^{\mathrm{st}}(\delta u, \delta u).$$

Applying (14) therefore yields (7) with $\kappa_\sigma^{\mathrm{sp}} = \kappa_\sigma^{\mathrm{st}}$. All requirements of Definition 2 are satisfied. □

6. Forced Constituents and Uniqueness

Proposition 2 (All spectral constituents are forced by the stationary Hessian law)**.** *Assume Definition 3. Then the following spectral-level constituents are fixed by the stationary Hessian law itself.*

1. *The spectral state body is necessarily the stationary state body $\mathcal{B}_\sigma^{\mathrm{sp}} = \mathcal{B}_\sigma^{\mathrm{st}}$.*
2. *The affine observer-data and shell-response readouts are necessarily $\Lambda_\sigma^{\mathrm{sp,dat}} = \Lambda_\sigma^{\mathrm{st,dat}}$ and $\Lambda_\sigma^{\mathrm{sp,sh}} = \Lambda_\sigma^{\mathrm{st,sh}}$.*

3. *The zero spectral band is necessarily the exact zero-hidden stationary locus*

$$V_\sigma^{\mathrm{sp},0} = V_\sigma^{\mathrm{st},0},$$

while the hidden spectral band is necessarily

$$H_\sigma^{\mathrm{sp}} = H_\sigma^{\mathrm{st}} = (V_\sigma^{\mathrm{st},0})^\perp.$$

4. *The spectral generator is necessarily the unique Riesz representative of the common zero-locus Hessian $\mathfrak{H}_\sigma^{\mathrm{st}}$, so its kernel, image, and hidden-band gap are all forced by that same stationary law.*

5. *The zero/hidden body splitting is necessarily the stationary splitting* (8)*, the hidden body is necessarily D_σ^{st}, and the orthogonal projector onto the hidden band is necessarily $\Pi_\sigma^{\mathrm{st,hid}}$.*

6. *The symmetry and time-reversal structure, the exact zero-response shell realization, fixed-data shell solvability on the zero band, the inert zero-hidden kernel and projector, and the spectral coercivity inequality are all induced by the same stationary law and are not additional spectral freedom.*

Proof. Items (1) and (2) are immediate from Definition 4. Item (3) is part of the stationary law itself. Item (4) is Lemma 1: once the common zero-locus Hessian is fixed, the spectral generator is uniquely determined by (15), and its kernel, image, and gap are forced. Item (5) is exactly the content of Definition 4: the zero/hidden splitting, hidden body, and hidden projector are the corresponding stationary objects and carry no further freedom. Item (6) is the content of Proposition 1: every remaining spectral constituent descends directly from the same stationary-Hessian package. □

Theorem 1 (Zero-hidden spectral generator law necessity / uniqueness from stationary Hessian law)**.** *Assume Definition 3. Then:*

1. *the canonical construction of Definition 4 yields a zero-hidden spectral generator law in the sense of Definition 2;*

2. *every spectral-level object singled out by the previous selector-law theorem is forced by the stationary law: the compact convex spectral state body, the affine observer-data and shell-response readouts, the positive self-adjoint spectral generator, the exact zero spectral band and positively gapped hidden spectral band, the induced zero/hidden splitting and hidden body, the symmetry and time-reversal structure, the exact zero-response shell realization on the zero band, fixed-data shell solvability on that locus, the inert zero-hidden kernel and projector, and the spectral coercivity inequality;*

3. *any other spectral package compatible with the same stationary-Hessian law differs from the canonical one only by the obvious orthogonal identification that preserves the zero/hidden splitting and the recorded data and shell coordinates;*

4. *consequently the spectral-law layer is not an independent remaining freedom once the stationary-Hessian law is fixed.*

Proof. Item (1) is Proposition 1. Item (2) is Proposition 2. For item (3), any compatible reparameterization must preserve the orthogonal decomposition

$$\mathcal{U}_\sigma^{\mathrm{st}} = V_\sigma^{\mathrm{st},0} \oplus H_\sigma^{\mathrm{st}},$$

must preserve the common Hessian $\mathfrak{H}_\sigma^{\mathrm{st}}$ and therefore its unique Riesz representative $\mathcal{L}_\sigma^{\mathrm{sp}}$, and must leave the recorded data and shell coordinates unchanged. That leaves only the obvious orthogonal identification respecting the zero/hidden splitting. Item (4) is the routing consequence of the previous items. □

7. Benchmark Preservation

Theorem 2 (The derived spectral law preserves the same one-metric benchmark package)**.** *Assume Definition 3. Then the benchmark output statements of Definition 1 remain unchanged:*

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi]$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*. Consequently the deeper stationary-Hessian law and the induced spectral law introduce neither a second operative metric nor an enlarged low-energy matter law.*

Proof. The present note changes only the upstream origin of the spectral data. It does not alter the primitive shell $\mathcal{S}_{\mathrm{prim}}$, the recorded metric map $\mathscr{G}_{\mathrm{out}}$, or the recorded matter map $\mathscr{T}_{\mathrm{out}}$. Those remain the fixed continuation data of Definition 1, so (2) continues to hold exactly. □

8. Main Theorem

Theorem 3 (The remaining burden moves below the spectral-law layer)**.** *Assume the fixed primitive continuation data of Definition 1 and, for each sector* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a zero-hidden stationary Hessian law in the sense of Definition 3. Then:*

1. *the zero-hidden spectral generator law of Definition 2 is canonically induced by the stationary-Hessian law;*
2. *the compact convex spectral state body, the affine observer-data and shell-response readouts, the positive self-adjoint spectral generator, the exact zero spectral band and positively gapped hidden spectral band, the induced zero/hidden splitting and hidden body, the symmetry and time-reversal structure, the exact zero-response shell realization on the zero band, fixed-data shell solvability on that locus, the inert zero-hidden kernel and projector, and the spectral coercivity inequality are all forced by that same stationary-Hessian law;*
3. *the one operative metric remains exactly* $g^\sharp$*, the ordinary matter route remains exactly the standard one through that metric, there is no second operative metric, and the low-energy matter law is not enlarged;*
4. *consequently the benchmark-facing burden after the previous selector-law theorem is discharged in the stronger form now available on the active line: the zero-hidden spectral generator law is no longer an unexplained stopping point, but a necessary reduction of a smaller zero-hidden stationary Hessian law.*

Proof. Item (1) is Theorem 1(1). Item (2) is Theorem 1(2). Item (3) is Theorem 2. Item (4) is the combined routing consequence. □

Corollary 1 (What remains open after this theorem)**.** *After Theorem 3, the remaining honest burden is no longer to justify the zero-hidden spectral generator law itself on the active primitive-observer-reduction line. The next honest burden is deeper again: derive or justify the zero-hidden stationary Hessian law from still more primitive substrate dynamics, or else prove a sharper inevitability theorem that removes even that remaining stationary-Hessian-level freedom while preserving the same benchmark one-metric package and claim boundary.*

Proof. Theorem 3 converts the spectral-law layer from a remaining benchmark-facing burden into a canonical reduction of a deeper stationary-Hessian law. What remains open is why that stationary-Hessian law itself should hold, rather than becoming the new disciplined stopping point. □

9. Conclusion

The positive result of this note is deliberately narrower than a completed final microphysical derivation and stronger than another deeper same-output relay. The previous theorem had already shown that the selector-law layer is a forced reduction of a spectral law. The present note goes one honest step further by showing that the spectral layer itself is not left hanging either. It is canonically induced by a smaller zero-hidden stationary Hessian law.

That gain directly addresses the precise objects that had become the next burden: the compact convex spectral state body, the affine observer-data and shell-response readouts, the positive self-adjoint spectral generator, the exact zero spectral band and positively gapped hidden spectral band, the induced zero/hidden splitting and hidden body, the symmetry and time-reversal structure, the exact zero-response shell realization on the zero band, fixed-data shell solvability on that locus, the inert zero-hidden kernel and projector, and the spectral coercivity inequality. Within the stationary-Hessian class those are no longer optional inserted ingredients. They are forced outputs, unique up to the harmless orthogonal identification preserving the zero/hidden splitting.

The benchmark boundary remains fixed throughout. The same primitive shell remains the shell carrying the observer outputs, so the one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. This is therefore a disciplined upstream derivational gain, not a final completion claim. The note still does not derive the stationary-Hessian law from ultimate substrate microphysics. That is the next honest open burden. But the spectral-law layer is no longer the stopping point on the active line.